



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Comparative NMPC Modeling for Tendon-Driven Continuum Robot Control

CREATE LAB SEMESTER PROJECT

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1 Introduction

Tendon-driven continuum manipulators are promising for applications requiring flexible motion, precise control, and safe interaction with delicate or unstructured environments, such as inspection and medical robotics [1], [2]. Unlike rigid-link robots, they deform continuously through tendon actuation, which makes their control nonlinear, underactuated, and strongly affected by friction, hysteresis, and dynamic coupling [3]. Nonlinear model predictive control (NMPC) is well suited to this because it predicts future motion while enforcing hardware and safety constraints, including tendon-tension bounds and tendon-rate limits [4]. However, real-time NMPC depends on prediction models that balance accuracy, computational efficiency, and hardware feasibility.

2 Background

Geometric and Mechanics-Based Models Continuum robots are often modelled using geometric or mechanics-based representations. Geometric models, such as piecewise constant curvature (PCC), provide a compact and physically interpretable description of the robot shape, making them attractive for real-time control and planning [1]. A schematic of the PCC representation used in this work is provided in Appendix B. However, basic PCC models often oversimplify the robot behaviour and may not capture nonlinear hardware effects such as tendon friction and hysteresis. More detailed mechanics-based models, such as Cosserat rod and finite-element formulations, can represent continuum deformation more accurately, but they typically require careful parameter identification and higher computational cost, which limits their use in fast closed-loop NMPC [5].

Data-Driven Dynamic Models Data-driven models learn robot dynamics directly from experimental input/output data. Approaches such as autoregressive models with exogenous inputs (ARX/NARX), recurrent neural networks, and sparse identification can capture nonlinear and history-dependent effects, where predictions depend on previous inputs and outputs [6], [7]. For tendon-driven continuum robots, these models can implicitly learn hardware dynamics. However, purely learned models rely on representative training data, may perform poorly on unseen motions, and can be harder to validate for safety-critical control.

Hybrid Models Recent work has moved toward hybrid or gray-box modelling approaches that combine physical structure with data-driven correction. These methods aim to preserve the physically meaningful structure and computational efficiency of model-based control while using data to compensate for dynamic effects that are difficult to model accurately [5], [8]. This makes them attractive for soft and continuum robots, where purely physical models can be oversimplified and purely learned models can be difficult to validate for safe control.

To address this trade-off, this report compares three prediction models within the same NMPC framework: **a PCC model, a learned NARX model, and an adaptive residual gray-box model**. The aim is to assess how model choice affects real-time tracking performance and whether a gray-box approach can balance accuracy, efficiency, and robustness.



Figure 1:
TDC
robot
platform.

3 Methods

NMPC Formulation At each control step, the controller solves a finite-horizon NMPC problem to track a desired bend-space trajectory while satisfying actuator constraints. All three controllers use the same outer NMPC structure but differ in their prediction model and internal state. Tracking is performed in bend space to avoid discontinuities in the bending-plane angle ϕ and bending magnitude θ ,

$$b = [b_x, b_y]^T = [\theta \cos \phi, \theta \sin \phi]^T.$$

The optimized input is the tendon command increment $\Delta T_k = [\Delta T_1, \Delta T_2, \Delta T_3]^T$, which updates the commanded tension as $T_{k+1} = T_k + \Delta T_k$.

The cost penalizes tracking error, bending velocity, tendon effort, and rapid command changes, with bounds on tendon tensions and increments. The first optimized command is applied in receding-horizon fashion.

Low-Level Current Control Motors are operated in current-control mode. NMPC tendon tensions are mapped into nominal motor currents

$$I_{\text{cmd},i} = I_{\text{nom},i} + I_{\text{fric},i}, \quad I_{\text{fric},i} = \hat{d} i a_i \left(I_{\text{dyn},i} + (I_{\text{stat},i} - I_{\text{dyn},i}) e^{-(|\omega_i|/\omega_c)^2} \right) \gamma_i.$$

This Stribeck-inspired friction compensation reduces near-zero-velocity stiction, which otherwise causes delayed staircase-like motion.

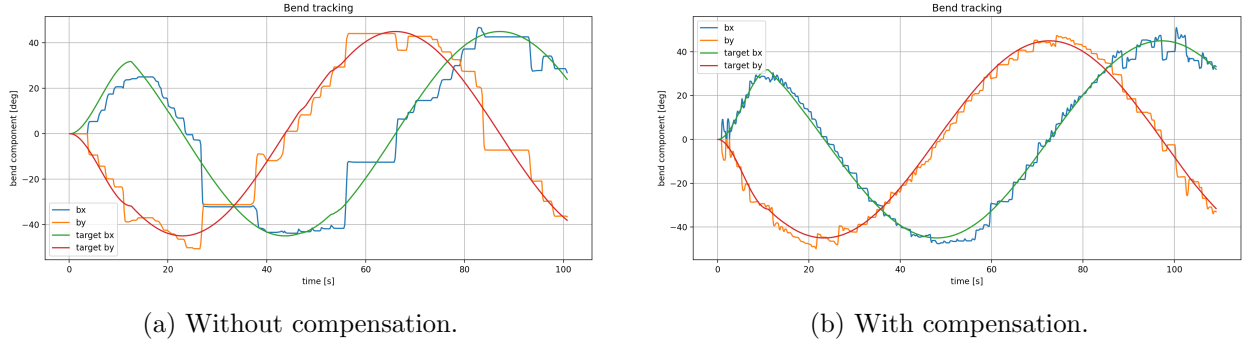


Figure 2: Effect of low-level friction compensation on bend tracking.

3.1 PCC-Based NMPC Prediction Model

The PCC-based model is used as the physics-based baseline and assumes that the continuum section deforms as a single constant-curvature segment parameterized by $q = [\phi, \theta]^T$. The corresponding backbone mapping and tendon Jacobian are given in Appendix B. The PCC dynamics are modeled in generalized coordinates as

$$M(q)\ddot{q} + C(q, \dot{q}) + D(q, \dot{q})\dot{q} + Kq + G(q) = J_T(q)^T T,$$

where $T = [T_1, T_2, T_3]^T$ is the tendon tension vector. The left-hand side captures inertia, velocity-dependent effects, damping, stiffness, and gravity, while $J_T(q)^T T$ maps tendon tensions into generalized bending forces. This model predicts the future bend state inside the NMPC, as summarized in Fig. 3.

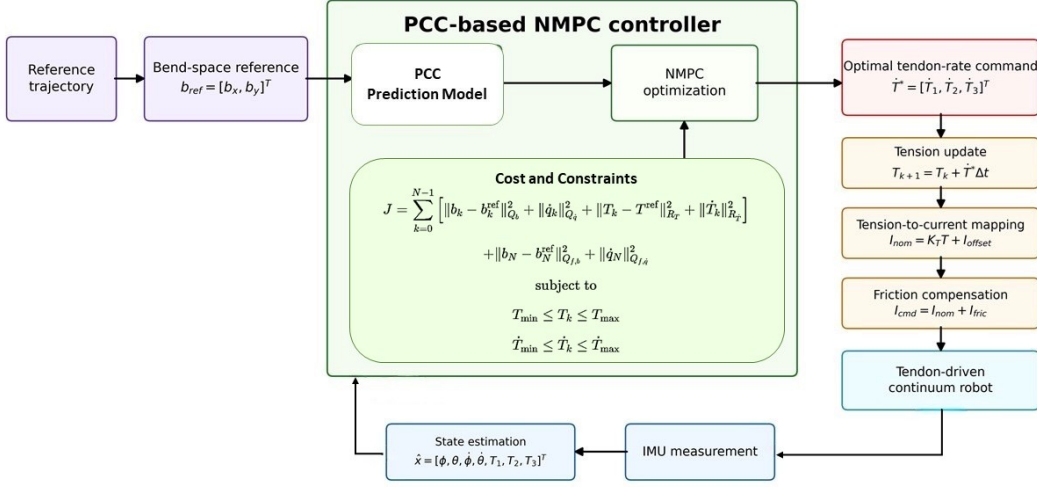


Figure 3: PCC-based NMPC control architecture.

3.2 Learned NARX-NMPC Model

The learned controller replaces the analytical prediction model with a delayed NARX model trained offline from open-loop hardware experiments, following the data-driven identification approach in [6]. PRBS, sinusoidal, and shape-excitation tendon commands were applied to collect bend-space outputs and tendon commands. The NARX model is trained in residual form, $y_{k+1} = y_k + \Delta \hat{y}_k$, using histories of bend outputs, tendon tensions, and tendon increments. Online, these histories define the augmented state $x_k = [y_k, \dots, y_{k-n_y}, T_k, \dots, T_{k-n_u}]^T$, which is updated after each IMU measurement and applied command. The workflow is shown in Fig. 4.

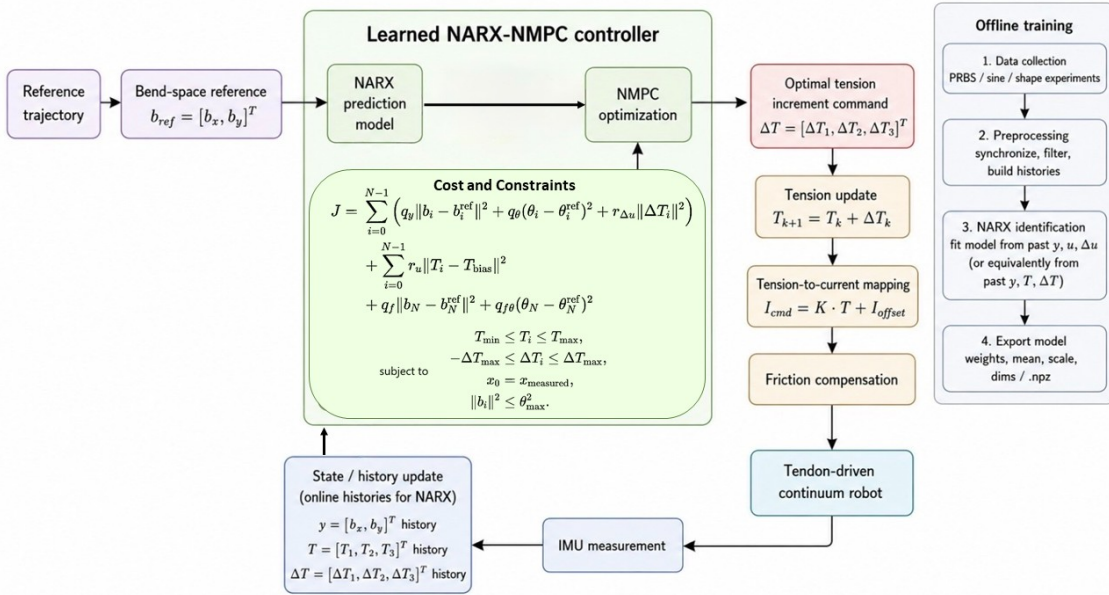


Figure 4: Learned NARX-based NMPC architecture.

3.3 Adaptive Residual Gray-Box Prediction Model

The adaptive residual gray-box model predicts bend-space acceleration and its prediction state is $x_k = [b_k^T \ v_k^T \ T_{\text{cmd},k}^T \ T_{\text{eff},k}^T \ d_k^T]^T$, where v_k is the bend velocity, T_{cmd} is the commanded tendon tension, T_{eff} is the delayed effective tension, and d_k is an online disturbance correction.

Since bending is mainly produced by differential tendon tension, the mean effective tension is removed and mapped to a planar bending moment:

$$\tilde{T}_{\text{eff},i} = T_{\text{eff},i} - \frac{1}{3} \sum_{j=1}^3 T_{\text{eff},j}, \quad m = r \sum_{i=1}^3 \tilde{T}_{\text{eff},i} \begin{bmatrix} \cos \beta_i \\ \sin \beta_i \end{bmatrix}.$$

The nominal acceleration model is

$$a_{\text{phys}} = W_{\text{phys}}^T f_{\text{phys}}, \quad f_{\text{phys}} = [1 \ b_x \ b_y \ v_x \ v_y \ m_x \ m_y]^T.$$

This feature vector is inspired by a second-order mechanical equation: the bend terms model stiffness-like behavior, the velocity terms model damping-like behavior, and the moment terms model tendon actuation. After fitting the nominal term from hardware data, the remaining acceleration error is learned as a residual correction a_{res} .

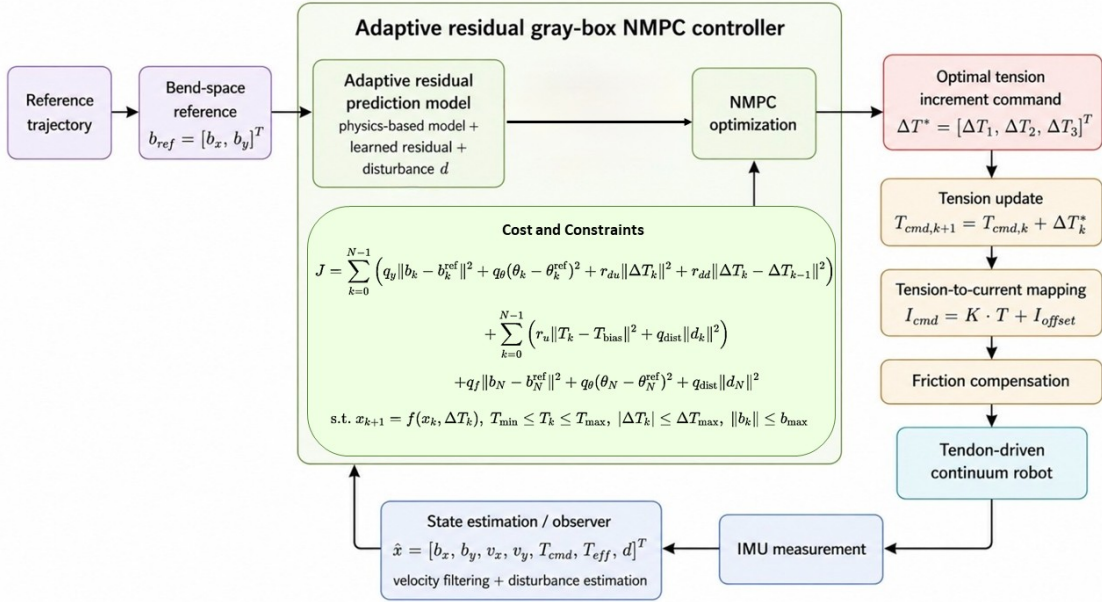


Figure 5: Adaptive residual gray-box NMPC control architecture.

The final predicted acceleration is then given by $a_{\text{pred}} = a_{\text{phys}} + a_{\text{res}} + d$, which is integrated forward to predict the bend state. The offline residual a_{res} captures repeatable modeling errors, while the observer updates d online after each IMU measurement to correct remaining slowly varying prediction mismatch, similarly to disturbance-observer-based compensation methods [9]. The complete workflow is shown in Fig. 5.

4 Experimental Setup

The experimental platform was a single TDC segment actuated by three Dynamixel XC330 motors, each driving one tendon. Robot bending was measured using a tip-mounted BNO055 IMU by computing the relative orientation with respect to a straight-reference pose and representing it in bend space. Three bend-space trajectories were tested: a circle, triangle, and star.

Table 1: Common settings used for all comparison experiments.

Parameter	Value
Sampling period	0.10 s (10 Hz)
Prediction horizon	$N = 10$ steps (1.0 s)
Tension limits	$0.35 \leq T_i \leq 4.0$ N
Tension increment limit	$ \Delta T_i \leq 0.35$ N/step
Reference bend magnitude	$\theta_{\text{ref}} = 50^\circ$

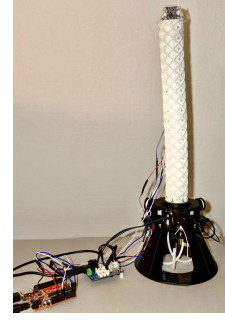


Figure 6: Setup

5 Results

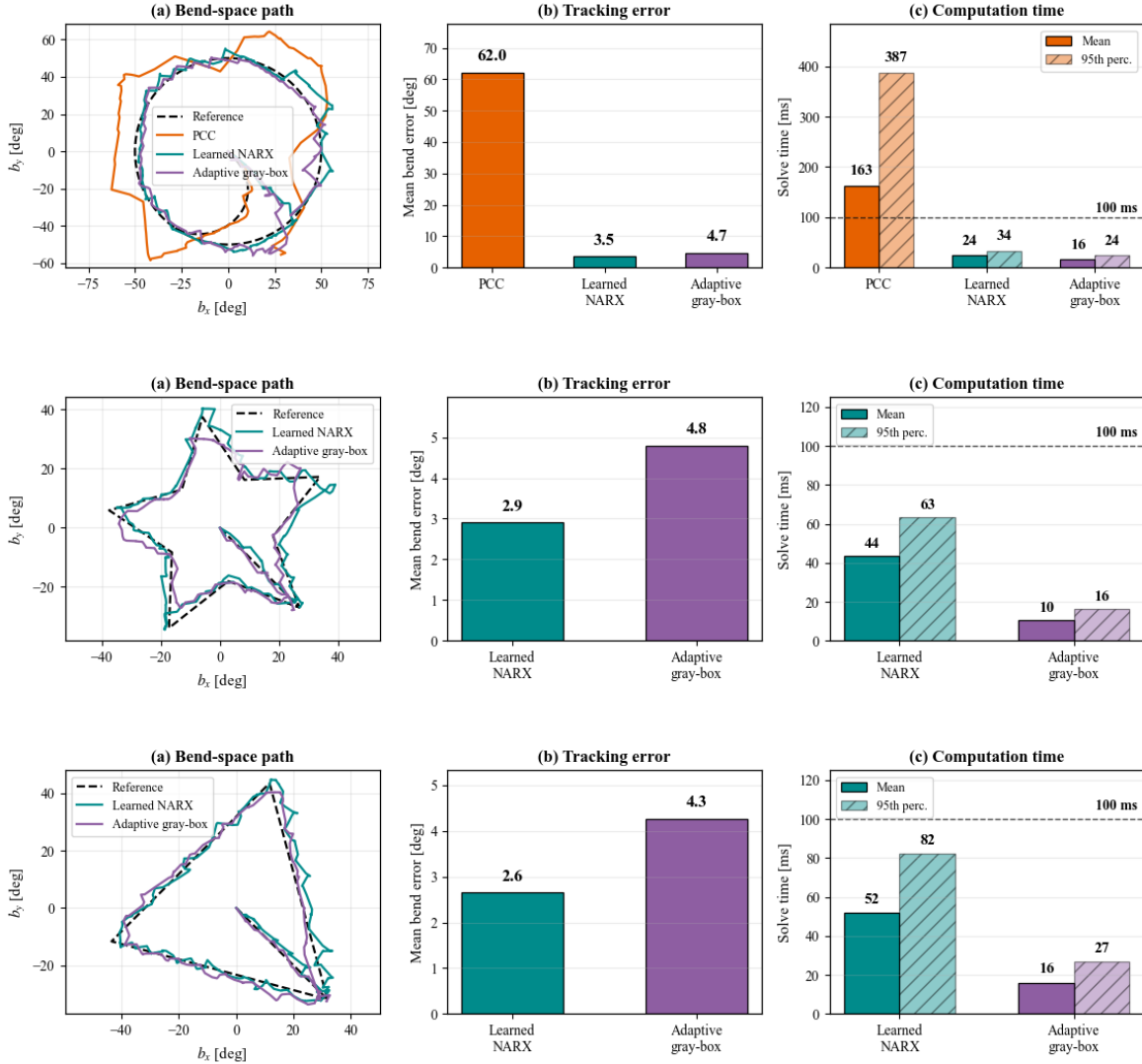


Figure 7: Trajectory tracking comparison for the circle, star, and triangle reference paths.

As shown in Fig. 7, the PCC-based NMPC shows a large mismatch from the reference path, with a much higher mean error than the learned and adaptive controllers, confirming that the simplified

physics-based model is not sufficient for the hardware behavior. Across the circle, star, and triangle trajectories, the learned NARX model gives the lowest tracking error, especially for the sharper star and triangle paths. However, the adaptive gray-box controller provides the best middle ground: its tracking is slightly less accurate than the learned model, but its computation time is much lower and more consistent, making it the most practical compromise for real-time NMPC.

Online Observer For the adaptive gray-box controller, the online observer compensates for slowly varying model mismatch during the experiment. In the star trajectory, the disturbance estimate in Fig. 8 varies over time, indicating correction of persistent effects such as friction, tendon slack, and hysteresis. This improves overall stability and shape tracking, although local errors remain near the sharp corners where the dynamics change rapidly.

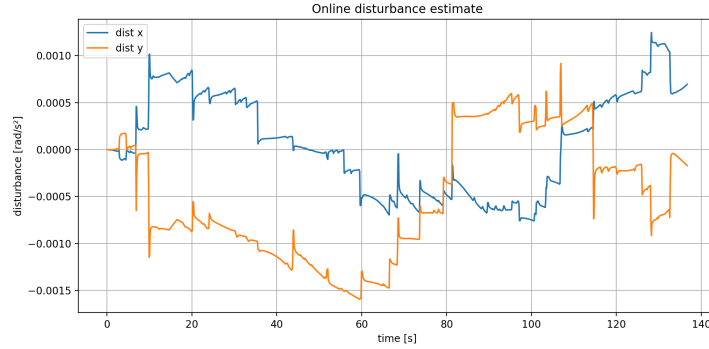


Figure 8: Online disturbance estimate of the adaptive residual NMPC during the star trajectory.

6 Discussion & Conclusion

This project compared PCC-based, learned NARX-based, and adaptive residual gray-box prediction models within the same NMPC framework for tendon-driven continuum robot control. The results show that model choice strongly affects closed-loop tracking. This performance trend suggests that accurate control cannot rely only on a simplified geometric model, since unmodeled hardware effects strongly influence the motion. At the same time, the comparison shows that adding physical structure to a learned correction can provide a useful middle ground: the adaptive gray-box model sacrifices some tracking accuracy relative to the fully learned model, but gains computational efficiency and more predictable real-time behavior. This trade-off may become increasingly important as the prediction horizon or model complexity increases.

A main **limitation** of the setup is that robot bending was estimated using only a single tip-mounted IMU, which was sufficient for bend-space tracking but did not provide direct tendon-force measurements. Since the robot was controlled in current mode, friction could affect the tendon tension, making friction compensation and tension estimation approximate without load sensors. Another **limitation** is that the experiments used a single-segment continuum robot, which is simplified compared with practical multi-segment manipulators. Extending the controllers to such systems would increase workspace and dexterity, but also introduce stronger dynamic coupling and more complex sensing.

Future work should therefore focus on extending the framework to multi-segment continuum manipulators, integrating tendon load cells or force sensors, improving friction and tension estimation, and testing faster and more complex trajectories. Further work should also explore hybrid learning approaches that combine data-driven accuracy with physics-based robustness and efficiency.

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Appendix

A Github Link

The complete source code used for data collection, model training, controller implementation, and experimental evaluation is available at:

[GitLab Repository](#)

B PCC Model Details

The PCC segment is described by the bending-plane direction ϕ , bending magnitude θ , and normalized arc coordinate $s \in [0, 1]$. For a segment of length L , the backbone position is

$$p(s, q) = Ls \begin{bmatrix} \cos \phi \operatorname{cosec}(s\theta) \\ \sin \phi \operatorname{cosec}(s\theta) \\ \operatorname{sinc}(s\theta) \end{bmatrix},$$

where

$$\operatorname{sinc}(z) = \frac{\sin z}{z}, \quad \operatorname{cosec}(z) = \frac{1 - \cos z}{z}.$$

For numerical stability near $\theta = 0$, these functions are evaluated using their small-angle series expansions.

The tendon actuation enters the generalized dynamics through the tendon Jacobian. For tendon k , with routing radius r_d and angular location σ_k , the corresponding row is

$$J_{T,k}(q) = [\theta r_d \sin(\sigma_k - \phi) \quad r_d \cos(\sigma_k - \phi)].$$

Stacking the three rows gives $J_T(q)$, which maps tendon tensions into generalized bending forces through $J_T(q)^T T$.

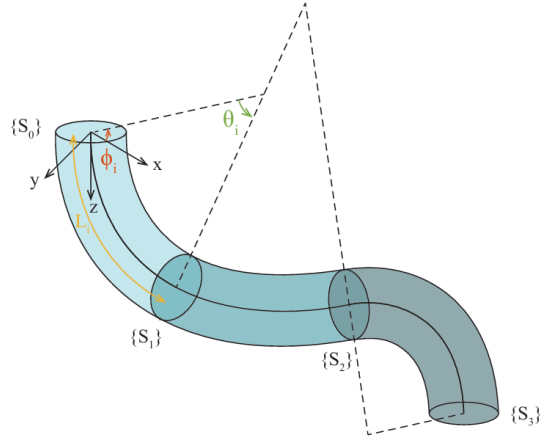


Figure 9: Piecewise constant curvature (PCC) parameterization of a continuum segment, showing the segment reference frames, bending-plane angle ϕ_i , bending angle θ_i , segment length L_i , and bending plane. Reproduced from [10].

C Results Diagnostics

C.1 Learned NARX-NMPC Star Trajectory Diagnostics

The learned NARX-NMPC controller tracked the overall star trajectory well, with a mean bend error of 2.91° and an RMS error of 3.27° . The largest errors occurred near the sharp transitions of the star, where the reference bend direction changes quickly and the measured bend response shows small delays and corner rounding. The tendon tensions remained within the imposed bounds, with a maximum value of 3.10 N, while the tendon-rate commands were mostly smooth and only briefly reached the ± 3.5 N/s limit. This suggests that the tracking errors were not mainly caused by persistent input saturation, but rather by model mismatch and hardware effects such as friction, slack, and release lag.

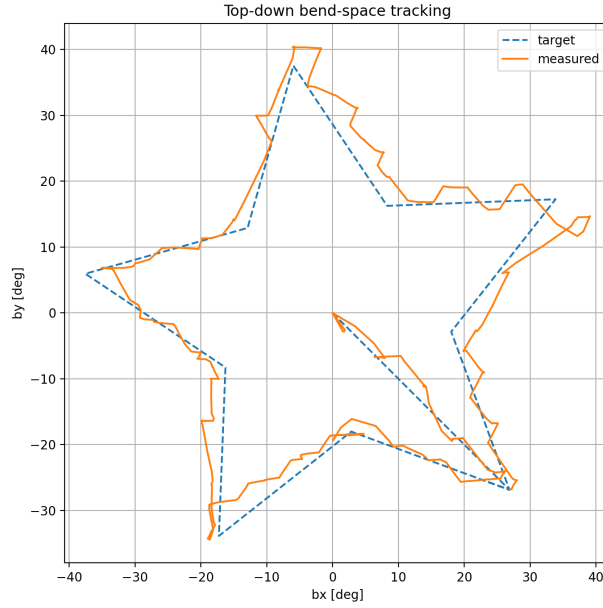


Figure 10: Learned Top-down Star View

Table 2: Learned NARX-NMPC performance metrics for the star trajectory.

Metric	Value
Mean bend error $[\circ]$	2.91
RMS bend error $[\circ]$	3.27
95th percentile bend error $[\circ]$	5.47
Maximum bend error $[\circ]$	9.72
Mean solve time [ms]	43.8
95th percentile solve time [ms]	63.3
Maximum solve time [ms]	451.0

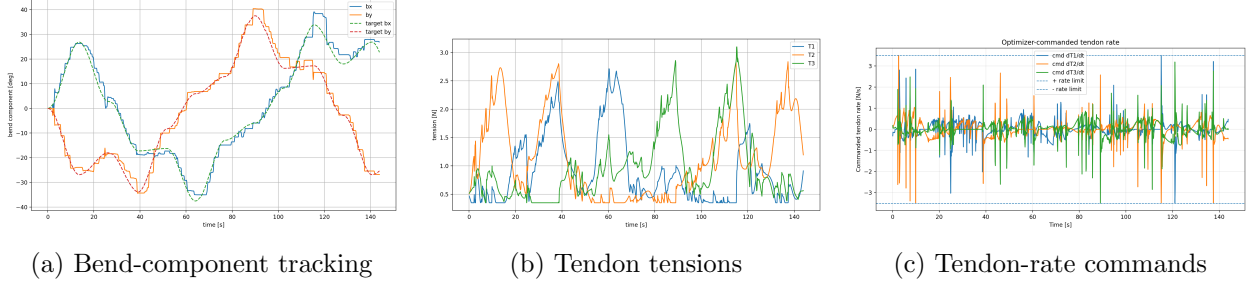


Figure 11: Learned NARX-NMPC diagnostics for the star trajectory.

C.2 Adaptive Gray-Box NMPC Star Trajectory Diagnostics

The adaptive gray-box NMPC controller follows the overall bend trend, but local delay and corner rounding remain visible near sharp transitions. The tendon tensions stay within bounds and the tendon-rate commands are not saturated, indicating that the remaining errors are mainly linked to model mismatch and hardware effects rather than actuator limits. The online observer estimates an additional disturbance term after each IMU measurement, which allows the prediction model to correct slowly varying mismatch. As shown by the disturbance estimate, the correction does not remain constant during the run, suggesting that the observer actively adapts to changing hardware behavior. However, it cannot fully remove the fast local errors at the star corners, where the dynamics change more quickly than the observer can compensate.

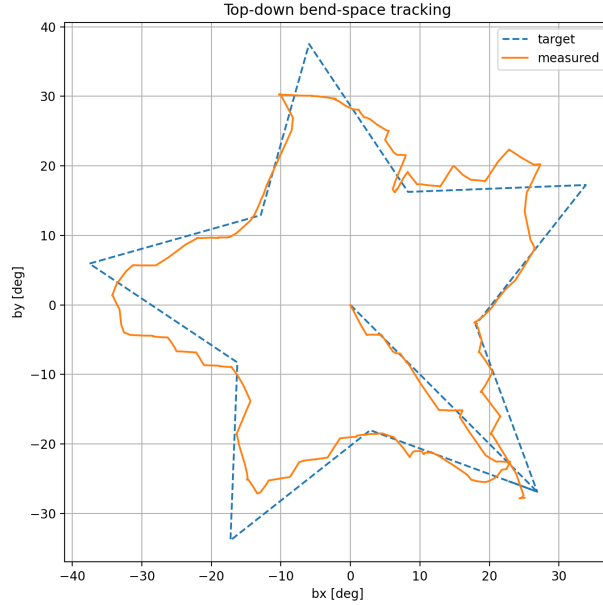


Figure 12: Adaptive Top-down Star View

Table 3: Adaptive gray-box NMPC performance metrics for the star trajectory.

Metric	Value
Mean bend error [$^{\circ}$]	4.79
RMS bend error [$^{\circ}$]	5.37
95th percentile bend error [$^{\circ}$]	8.64
Maximum bend error [$^{\circ}$]	11.67
Mean solve time [ms]	10.4
95th percentile solve time [ms]	16.3
Maximum solve time [ms]	29.1

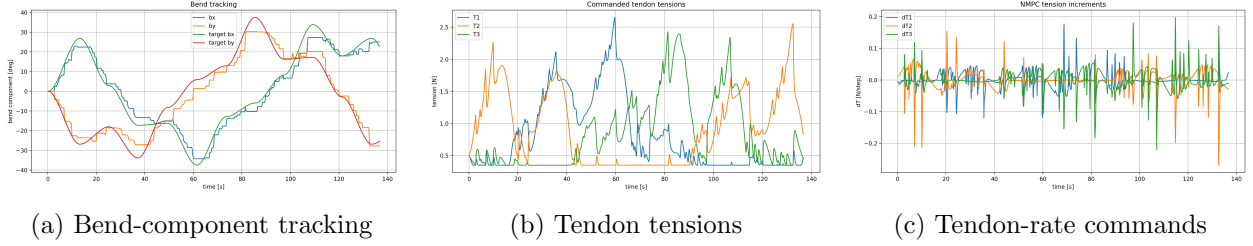


Figure 13: Adaptive gray-box NMPC diagnostics for the star trajectory.

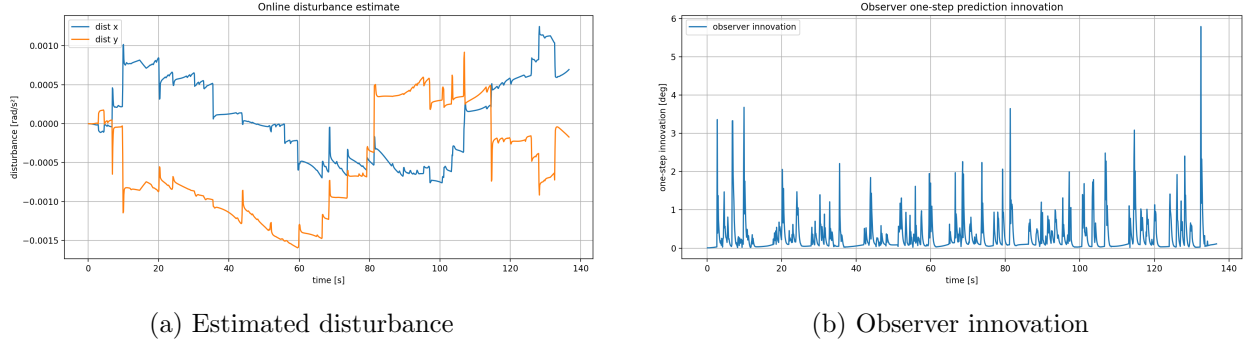


Figure 14: Observer diagnostics for the adaptive gray-box NMPC during the star trajectory. The disturbance estimate varies over time, indicating online compensation of slowly changing model mismatch, while the innovation highlights the remaining prediction error used to update the observer.